

Honors Intro Micro | In Class

November 18, 2021

Vignette 9 | Wandmaker's Dilemma

Suppose the demand for wands is given by

$$P = 200 - Q$$

and there are two main wand sellers, Olivander and Gregorovitch, that can make wands at a constant marginal cost of 20. The marginal revenue for Olivander is

$$MR_O = 200 - 2q_O - q_G$$

and for Gregorovitch it is

$$MR_G = 200 - 2q_G - q_O$$

Units are in galleons and stones. Calculate the Nash equilibrium level of output for the two wandsellers and put it in the blanks below.

Part A. Olivander's Quantity

What is the Nash equilibrium quantity for Olivander?

Part B. Gregorovitch's Quantity

What is the Nash equilibrium quantity for Gregorovitch?

Part C. Equilibrium Price

What is the Nash equilibrium price in this market?